

FlowPilot: A Multi-Agent AI System for Continuous Flow Chemistry Process Design

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ABSTRACT

Designing a continuous flow process demands coordinated reasoning across fluid dynamics, heat and mass transfer, materials compatibility, and safety, a multi-domain problem exceeding the reliable scope of a single-shot large language model (LLM). We present FlowPilot, to our knowledge the first multi-agent AI pipeline for sustainable continuous flow process design. LLM agents interpret chemistry and mechanism, while a nine-step deterministic calculator and approximately 2,500 handbook-derived design rules perform all quantitative computation. A seven-agent Council deliberates over candidate designs under bounded domain authority, selecting a final design via a weighted Process Value Score quantifying improvement over batch. Benchmarking across a 4×4 matrix of commercial and local LLM endpoints shows that the council contributes measurable design quality beyond model capability alone. We validate FlowPilot on an isoxazole cycloaddition benchmark and two end-to-end case studies: a photoredox Giese addition with aerobic oxidation and a thermal α -bromination with inline quench.

INTRODUCTION

Continuous flow chemistry has matured into one of the main technologies^{1,2} for pharmaceutical³, fine-chemical⁴, and on-demand manufacturing⁵, offering orders-of-magnitude improvements in heat and mass transfer⁶, intrinsically safer handling of hazardous intermediates⁷, access to short residence times², and ready accommodation of photochemistry⁸, electrochemistry⁹, liquid-phase oxidation¹⁰, and multi-step coupling^{11–13}. These intensified conditions translate directly into reduced solvent consumption, lower energy input, and smaller equipment footprints, aligning flow synthesis with the resource-efficiency goals of sustainable manufacturing. A mature commercial ecosystem now supplies virtually everything a flow chemist needs¹⁴, yet adoption remains gated less by what flow can do than by what it costs to design: the gap between a published batch protocol and a working flow process is typically closed by a small team over weeks of iteration; even within established translation paradigms such as microwave-to-flow, each

new chemistry class typically requires re-derivation of operating conditions¹⁵. The bottleneck has shifted from running flow chemistry to designing flow processes.

Designing a flow process is a chain of interlocking decisions across fluid dynamics^{16,17}, mass transfer¹⁸, heat transfer¹⁹, materials compatibility, kinetics, and safety, where each decision constrains the next^{20,21}. This coupled, multi-domain design surface is forbidding for an individual chemist. Existing tools address fragments of it: classical process simulators²² solve the engineering rigorously but require a pre-specified flowsheet, and reinforcement-learning approaches relax this restriction only under simplified process abstractions²³; Bayesian and active-learning optimizers^{24–28} refine a known process but cannot propose one; retrosynthesis tools^{29–31} answer what to make but not how to run it in flow. Handbook knowledge^{20,21,32} encodes the rules of thumb an experienced flow chemist relies on, but this knowledge is dispersed across textbook chapters, qualitative in form, and almost never operationalized computationally. No current tool designs a flow process from a chemistry brief while honoring these handbook-level fundamentals.

Recent large-language-model (LLM) copilots for chemistry^{33,34} have expanded what AI can do for a chemist, from synthesis planning to autonomous experimentation^{35–38}, but these efforts remain largely batch-oriented. When a monolithic LLM is prompted directly for a flow reactor design, it must handle chemistry, kinetics, fluid mechanics, heat transfer, and safety within a single inference pass^{39,40}, with no mechanism to enforce cross-domain consistency. The predictable outcome is internally inconsistent outputs and no traceable audit of which constraints were evaluated.

The structure of flow process design, a coupled problem solved by a team of complementary specialists, maps naturally onto a multi-agent architecture in which each agent owns one domain and a separate orchestrator integrates their judgments. The organizing principle of this work is therefore: LLMs interpret chemistry; deterministic code, anchored in handbook fundamentals, computes.

We introduce FlowPilot, to our knowledge the first multi-agent AI pipeline purpose-built for continuous flow chemistry process design. Given a batch protocol or a target reaction, FlowPilot returns a defensible, intensification-aware flow process design accompanied by a process flow diagram, traceable engineering calculations, mechanism-matched literature analogies, and explicit uncertainty flags. The system comprises five coupled stages: (i) a chemistry-interpretation agent that produces a structured plan including an IntensificationMandate, a computable target that quantifies the minimum residence-time reduction, throughput gain, and intensification factor a flow design must deliver over the published batch protocol; (ii) plan-aware retrieval-augmented generation (RAG) against a curated flow-chemistry corpus,

in which a chemistry-conditioned query expansion replaces single-pass semantic search; (iii) a nine-step deterministic engineering calculator grounded in approximately 2,500 handbook-derived design rules; (iv) a seven-agent Council that audits candidate designs under bounded domain authority and selects a final design through a weighted, internally computed score including an explicit Process Value Score (used only by the orchestrator); and (v) automatic rendering of a chemistry-aware process flow diagram.

This work makes the following contributions. We present the FlowPilot architecture and its hybrid LLM-deterministic design. We introduce a computable IntensificationMandate that converts the qualitative question of whether flow improves on batch into a quantitative design gate. We benchmark FlowPilot on a 4×4 matrix of upstream and council models spanning commercial endpoints (Claude Sonnet, GPT-4o, GPT-4o-mini) and a local model (Gemma 4 31B), yielding 16 completed pairings including lightweight modes, and demonstrate through pre/post-council comparison that the council contributes measurable design quality beyond model capability alone. Finally, we validate the system on an isoxazole 1,3-dipolar cycloaddition benchmark⁴¹ and two end-to-end case studies, a photoredox Giese addition with downstream aerobic oxidation⁴² and a thermal α -bromination with inline quench⁴³.

RESULTS

FlowPilot was designed to translate a batch chemistry protocol into a continuous-flow process proposal while keeping chemical interpretation, deterministic engineering calculation, and expert review as separate operations (Figure 1). The input layer accepts either a user query, a batch protocol, a chemistry goal, or a laboratory inventory. The upstream module then converts this unstructured input into a chemistry plan, including species identification, reaction classification, stream-separation logic, sensitivity flags, and an explicit statement of why flow should offer value over batch. This chemistry plan is passed to a deterministic design calculator that generates the initial engineering proposal, including residence time, reactor geometry, stream assignments, and operating conditions.

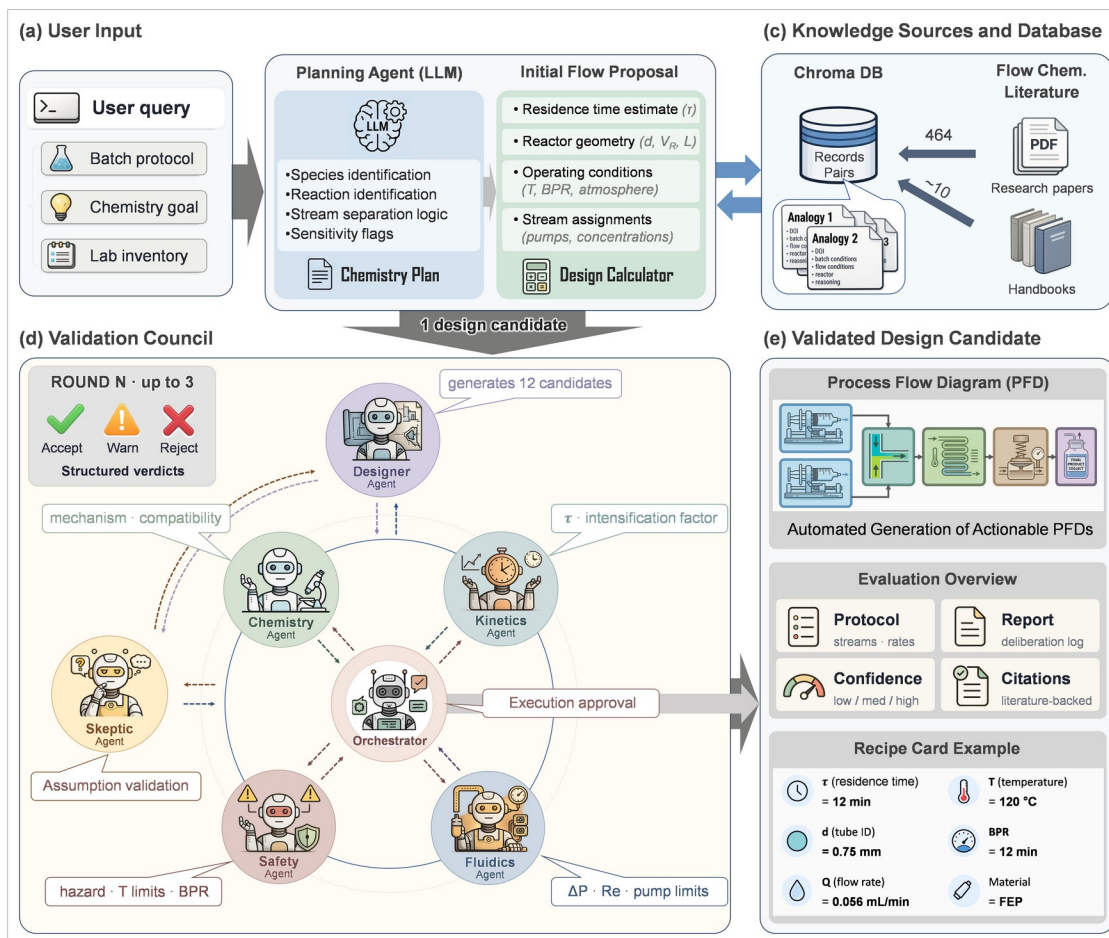


Figure 1. FlowPilot system architecture. Input layer accepts batch protocols, user queries, chemistry goals, or laboratory inventories. The upstream module produces a structured chemistry plan with an IntensificationMandate; the nine-step deterministic engineering calculator generates initial designs; the seven-agent Council deliberates over candidates; and the topology builder renders a chemistry-aware process flow diagram.

The downstream council provides the main error-control mechanism. Rather than asking one model to produce a single final answer, the system generates a candidate pool and distributes review across domain-specialized agents. The Designer proposes candidate geometries and operating points; Chemistry Agent evaluates chemical compatibility, photochemical constraints, and stream logic; Kinetics Agent evaluates residence time and conversion plausibility; Fluidics Agent evaluates flow regime, pressure drop, mixing, and hardware compatibility; Safety Agent evaluates pressure, thermal, material, and handling risks; the Skeptic audits assumptions and internal consistency; and the Orchestrator selects the final

candidate. This structure produces not only a selected design but also a record of rejected alternatives, revised candidates, disqualified candidates, and the rationale for each decision.

The output of the workflow is a validated design package rather than only a text answer. The final response includes a process-flow diagram, an evaluation overview, protocol and report summaries, confidence and safety flags, and citations or retrieval-backed analogies where available. The system therefore exposes both the final operating point and the path by which that point was selected. Detailed architecture, authority boundaries, intake contracts, and run provenance are provided in Figures S1-S2 and Tables S1-S3.

Literature and rule-base coverage

The literature dataset used for retrieval and knowledge extraction covers a broad range of flow-chemistry applications (Figure 2). The curated corpus contained annotated query–record pairs drawn from 464 papers spanning thermal synthesis, photoredox catalysis, heterogeneous photocatalysis, cross-coupling, oxidation/reduction, and other reaction classes. Reactor types ranged from capillary/coil and microreactor/chip systems to photoreactors and packed-bed reactors, with fluoropolymer tubing (PTFE, PFA, FEP) dominant among materials. Most annotated processes used two or three inlet streams, supporting the system’s default topology assumptions. This diversity matters because the benchmark and case-study reactions include both photochemical and non-photochemical processes, and the retrieval stage must therefore recognize different reactor classes, materials, and operating regimes.



Figure 2. Literature corpus and engineering rule base coverage. (a–d) Distribution of curated query–record pairs drawn from 464 flow-chemistry papers across reaction classes, reactor types, materials, and inlet-stream configurations. (e–f) The structured engineering rule base contains 2,537 rules spanning reactor design, heat transfer, catalyst behavior, residence time, photochemistry, materials, scale-up, pressure, mixing, and safety.

The engineering knowledge base contained 2,537 structured rules spanning reactor design, heat transfer, catalyst behavior, residence time, photochemistry, materials, scale-up, pressure, mixing, and safety (Figure 2e–f). Coverage was not uniform across chemistry classes: heat-transfer rules were densely associated with thermal synthesis and cross-coupling, whereas photochemistry rules concentrated in photoredox and photocycloaddition classes. This structured rule landscape constrains model-generated

designs with flow-chemistry fundamentals instead of relying only on model priors. The frozen corpus composition and computed rule-base coverage are reported in Figures S3-S4 and Tables S4-S5.

Plan-aware retrieval improves analogy quality

The retrieval system was evaluated against a standard semantic-search baseline (Figure 3). A standard RAG pipeline embeds the raw batch protocol or query summary and ranks records by cosine similarity. FlowPilot instead expands the query using the chemistry plan, including reaction class, photocatalyst family, solvent, wavelength, and mechanistic constraints. Candidate records are retrieved through a three-tier filtering strategy (Figure 3a): the first tier applies hard metadata filters (e.g., reaction class), the second relaxes these filters to broaden coverage, and the third falls back to pure semantic retrieval when no metadata match is found. Surviving candidates are then reranked by a weighted field score that rewards mechanistically relevant matches.

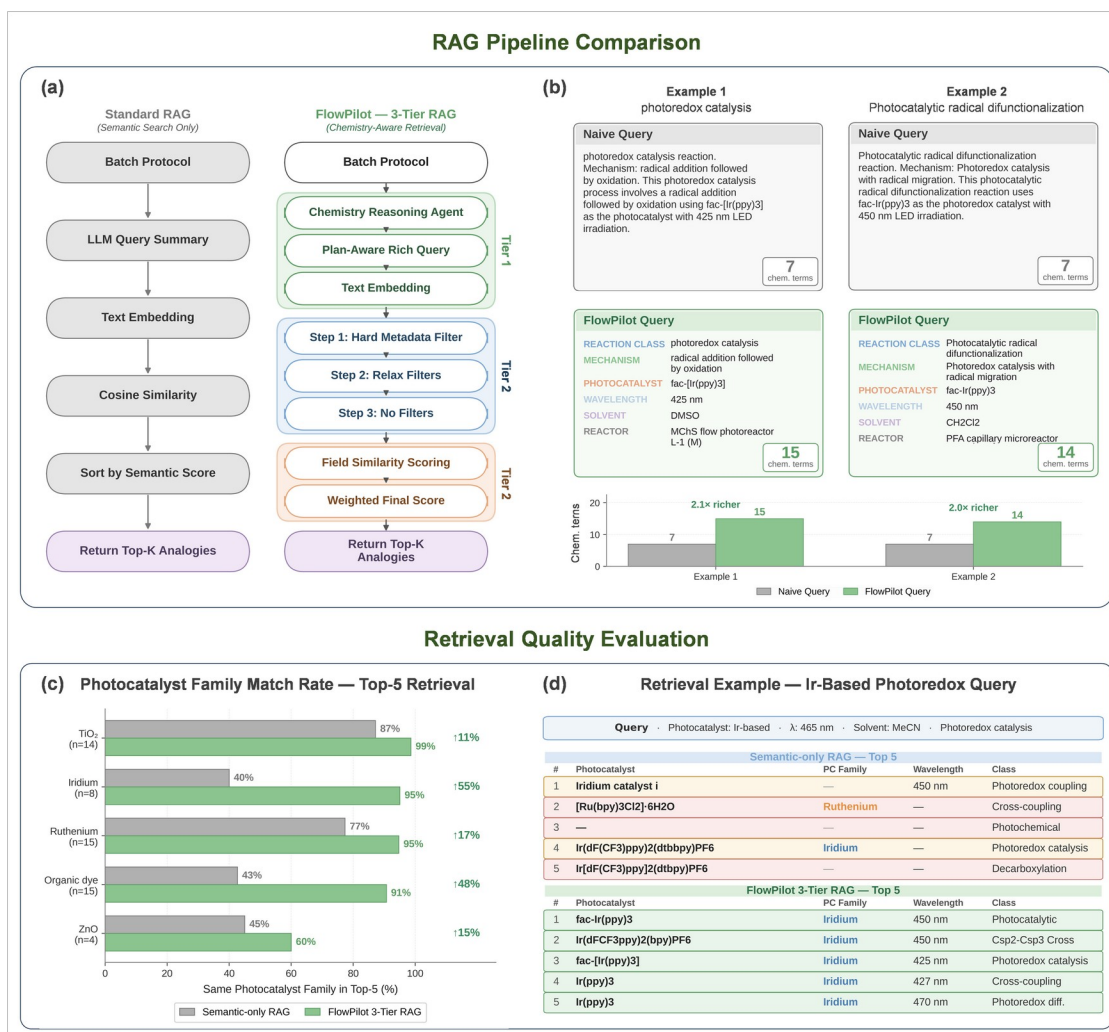


Figure 3. Plan-aware retrieval. (a) Three-tier filtering strategy: hard metadata filter, relaxed metadata filter, and pure semantic fallback. (b) Query enrichment expands chemically specific search terms approximately two-fold relative to a naive semantic baseline. (c) Top-5 photocatalyst-family matching rates by family (iridium, organic dye, ruthenium, TiO₂, ZnO) for semantic baseline vs FlowPilot reranking. (d) Representative reranking outcome showing closer wavelength and photocatalyst-family alignment.

The enriched query representation approximately doubled the number of chemically specific search terms in the two examples shown in Figure 3b. A photoredox-catalysis query increased from 7 naive terms to 15 FlowPilot terms, and a photocatalytic radical difunctionalization query increased from 7 to 14 terms. In the retrieval-pair benchmark, 1,600 query–result pairs were evaluated. FlowPilot reranked 25.1% of these

1,600 pairs relative to the pure semantic baseline, indicating that field-aware scoring materially changed which analogies were presented to the design engine.

The strongest retrieval improvement was observed for photocatalyst-family matching. For iridium systems, top-5 family matching increased from 40.0% under semantic search to 95.0% under FlowPilot reranking, a 55 percentage-point improvement. Organic-dye systems improved from 42.7% to 90.7%, ruthenium systems from 77.3% to 94.7%, TiO₂ systems from 87.1% to 98.6%, and ZnO systems from 45.0% to 60.0%. The demonstration query in Figure 3d illustrates the mechanism: semantic search retrieved superficially similar photochemical examples, whereas FlowPilot preferentially promoted records with the same photocatalyst family and closer wavelength context. This matters for downstream design because residence time, optical path length, material selection, and LED wavelength are not interchangeable across photoredox families. Retrieval tiers, scoring weights, benchmark data, and leakage controls are detailed in Figures S5-S6 and Tables S6-S7.

Ablation and architecture comparison

FlowPilot was evaluated through matched architecture comparisons that held the chemistry, inventory, and requested output contract constant while varying whether the generator operated as a one-shot model or within the full pipeline (Figure 4). The benchmark combined universal fixed criteria, deterministic critical-error checks, and independent LLM judges. Three chemistry cases and three generation repeats were used for each retained model-architecture pairing.

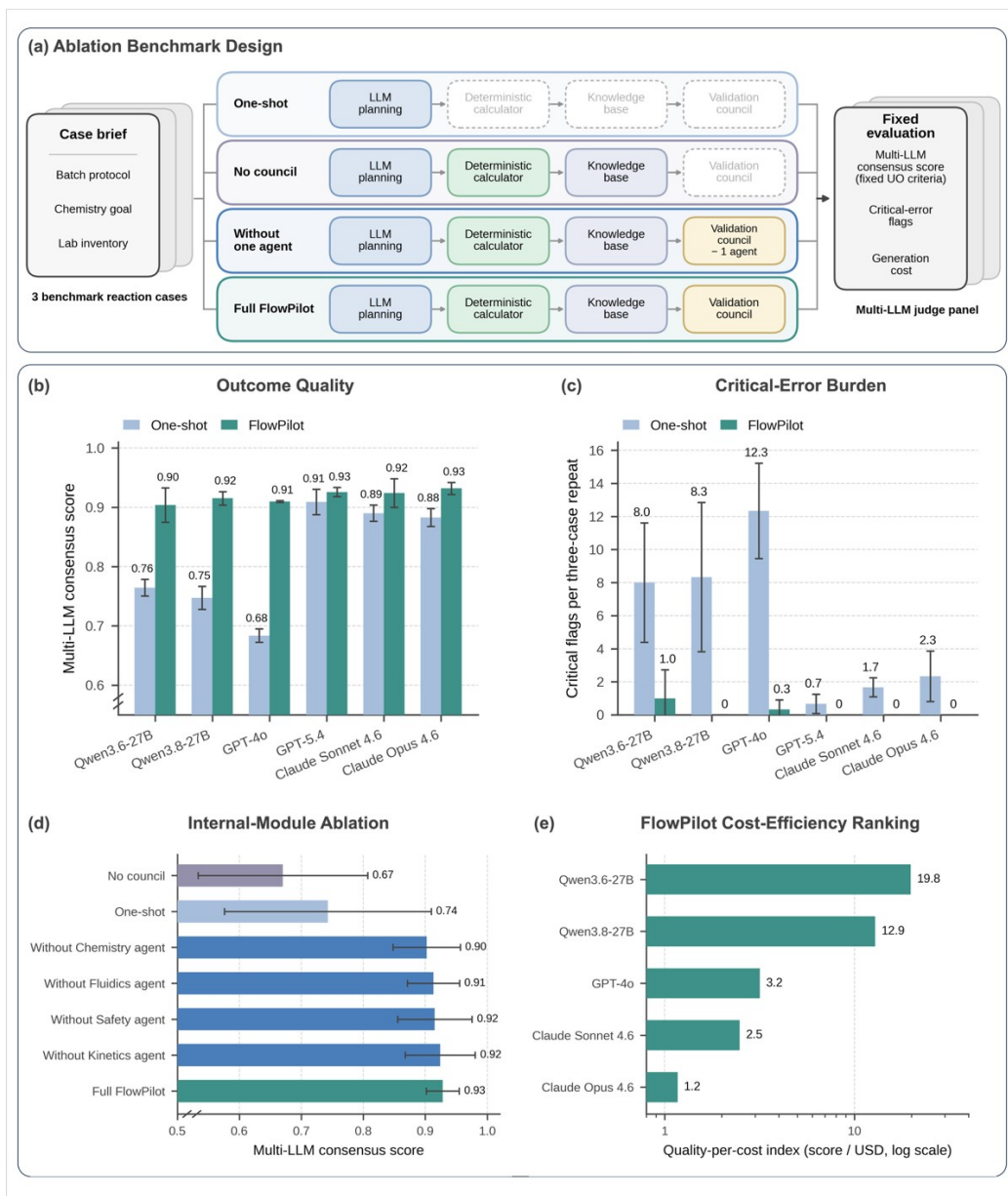


Figure 4. Ablation benchmarking of FlowPilot performance and cost efficiency. (a) Benchmark design comparing four configurations—One-shot, No council, Without one agent, and Full FlowPilot—across three reaction cases using the same inputs and fixed external evaluation criteria. (b) Multi-LLM consensus scores for One-shot and Full FlowPilot across six models, showing consistently improved outcome quality with FlowPilot. (c) Critical-error flags per three-case repeat, demonstrating a strong reduction in critical errors with FlowPilot relative to One-shot generation. (d) Internal-module ablation showing the effect of removing the council or individual specialist agents on consensus score; removal of the full council produces the largest performance loss. (e) FlowPilot quality-per-cost ranking, calculated

as benchmark score divided by generation cost (USD, log scale), highlighting the higher cost efficiency of the Qwen models relative to the commercial models. Overall, the results show that deterministic calculation and multi-agent validation improve design quality while reducing critical errors.

Across the retained benchmark campaigns, the full pipeline generally increased the fixed-criteria benchmark score and reduced campaign-level critical errors relative to matched one-shot generation. Internal ablations showed that removal of the council produced the clearest loss in error control, while specialist-agent removal produced smaller, chemistry-dependent changes. Cost and latency therefore represent the price of explicit decomposition, recomputation, and review rather than an unmeasured increase in prose length.

Supporting benchmark results are organized in their order of citation: Figures S7-S10 and Tables S8-S12 provide scoring anchors, universal criteria, ablations, matched architecture scores, critical-error maps, and resource use; Tables S13-S16 define deterministic engineering closure, inventory realization, experimental feedback, and computational replay. Figure S11 and Table S17 quantify the matched user-input and returned-output burden.

The earlier council model-matrix and candidate-budget experiment has been moved to the ESI (Figure S12 and Tables S18-S21). In the five-repeat comparison, the normalized radar-area score changed from 0.31 +/- 0.03 before council review to 0.48 +/- 0.23 after review, rather than the earlier single-run description based on 0.29 to 0.83. The ESI also defines Need revision, New versions, Combined pool, and Disqualified and reports the complete runtime, token, geometry, budget, and radar-normalization data. Figure S13 then compares the executable FlowPilot topology with the one-shot structured response.

Case study 1: photoredox Giese addition followed by aerobic oxidation

The first end-to-end case study was a one-pot photoredox Giese addition followed by aerobic oxidation to a sulfoxide product⁴² (Figure 5). The batch protocol required strict oxygen exclusion in the first step, followed by oxygen introduction in the second step. This presents a topology challenge: the system must preserve an anaerobic photoredox radical-addition stage while enabling an aerobic oxidation stage downstream.

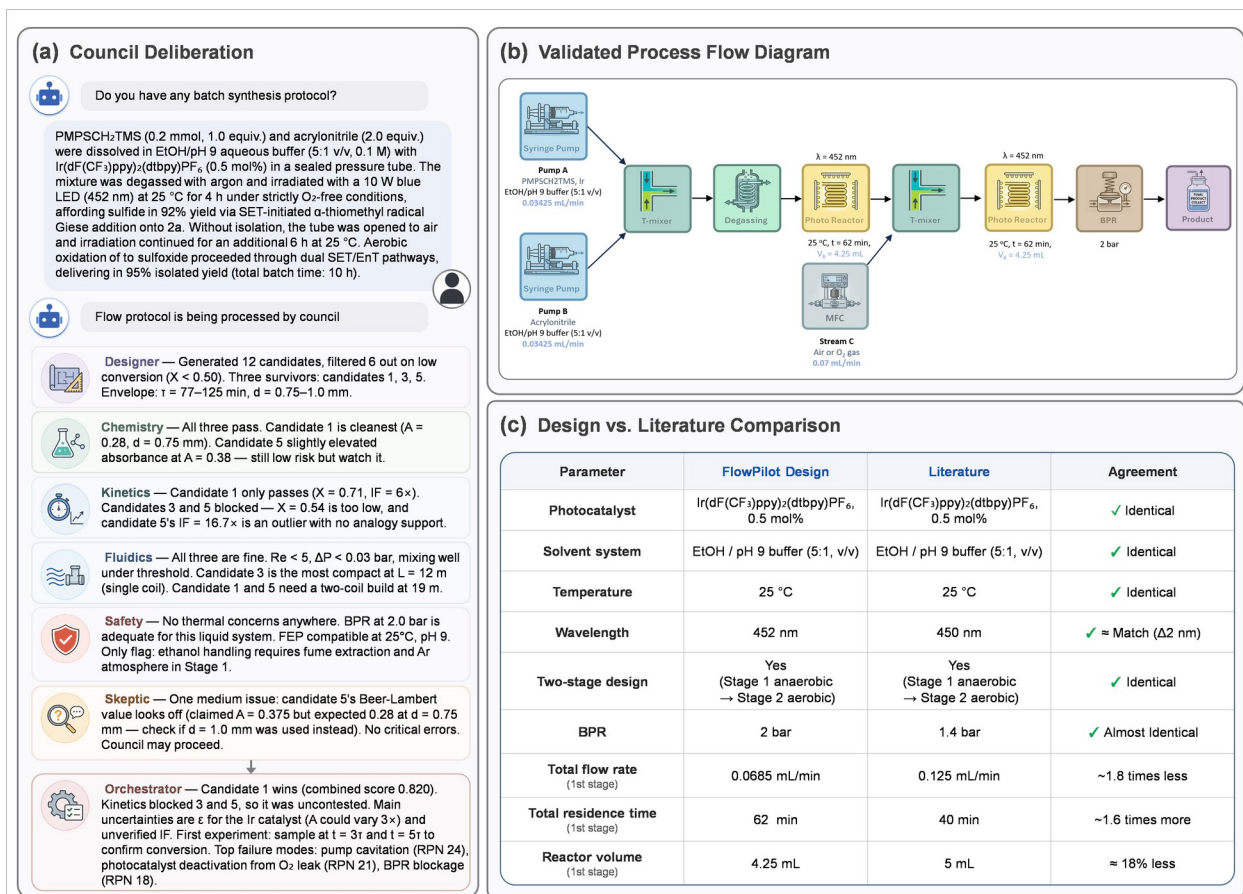


Figure 5. Case Study 1 — photoredox Giese addition with downstream aerobic oxidation. (a) Excerpts of council deliberation log. (b) FlowPilot-validated process flow diagram with anaerobic photoreactor stage and aerobic oxidation stage. (c) Parameter-by-parameter comparison between the FlowPilot design and the published flow protocol; the quantitative comparison is restricted to Stage 1 (single-phase liquid).

FlowPilot correctly identified the need for a staged process. The generated process-flow diagram contained two liquid feeds, an initial mixer, a degassing step, a first photoreactor, an MFC-controlled air or oxygen feed into a second mixer, a second photoreactor, a back-pressure regulator, and product collection. The system preserved the literature photocatalyst, solvent system, temperature, wavelength, and two-stage logic. The design used Ir(dF(CF₃)ppy)₂(dtbpy)PF₆ at 0.5 mol%, EtOH/pH 9 buffer (5:1 v/v), 25 °C operation, and 452 nm irradiation, matching the reported 450 nm literature wavelength to within 2 nm.

The council deliberation exposed the trade-off that made this case difficult. The Designer generated 12 candidates, filtered out six low-conversion candidates, and retained three survivors. Chemistry Agent judged all three chemically plausible, noting that Beer-Lambert absorbance remained low risk. Kinetics

Agent was more restrictive and accepted only one candidate; other candidates were blocked for insufficient estimated conversion or unsupported intensification. Fluidics Agent found all three candidates acceptable from pressure and Reynolds-number perspectives, but noted coil-length differences. Safety Agent found no major thermal concern and accepted 2 bar BPR operation for the liquid system, while flagging ethanol handling and the need to maintain the inert first stage. The Skeptic identified a medium-level inconsistency in one Beer-Lambert estimate but no critical error. The Orchestrator selected Candidate 1 with a combined score of 0.820.

Compared with the literature process, FlowPilot reproduced the qualitative process structure but selected a more conservative hydraulic point. The final design used a liquid flow rate of 0.0685 mL/min, total residence time of 124 min, and total reactor volume of 8.5 mL (two stages of 4.25 mL each). Because the current deterministic calculator computes residence times on a liquid-phase basis and does not account for gas-phase holdup in the aerobic second stage, we restrict the quantitative comparison to Stage 1, which is a single-phase liquid system in both the FlowPilot design and the literature process. For Stage 1, FlowPilot selected a residence time of 62 min and a reactor volume of 4.25 mL at a flow rate of 0.0685 mL/min, compared with the literature values of 40 min and 5 mL at 0.125 mL/min⁴². The system therefore overestimated the Stage 1 residence time by approximately 1.6-fold and operated at a flow rate approximately 1.8-fold lower, while proposing a reactor volume approximately 15% smaller. The literature process additionally reports a 1.4 bar BPR versus the 2 bar selected by FlowPilot. This case illustrates both the strength and limitation of the current system: it can infer a chemically correct staged photochemical topology, but kinetic uncertainty in the absence of experimentally measured rate constants drives conservative residence-time selection. Extending the calculator to handle gas–liquid multiphase residence times is an identified direction for future work.

Case study 2: thermal alpha-bromination with inline quench

The second case study was thermal alpha-bromination of 3-phenylpropanal with bromine in MeCN, followed by sodium thiosulfate quench⁴³ (Figure 6). The batch protocol used 3-phenylpropanal and Br₂ at 25 °C under N₂, followed by aqueous Na₂S₂O₃ quench and workup. The key design challenge was fast mixing and controlled bromine addition rather than light delivery to a photocatalyst.

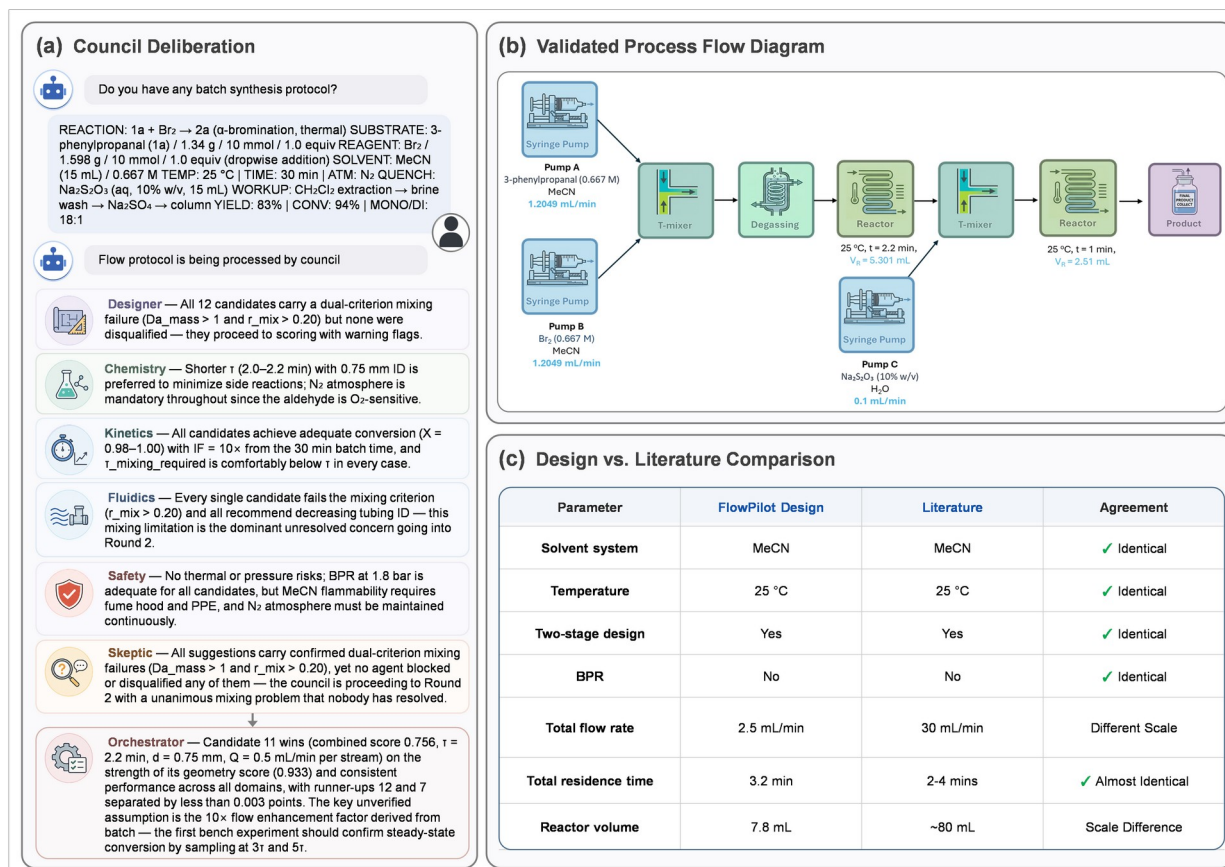


Figure 6. Case Study 2 — thermal α -bromination of 3-phenylpropanal with inline sodium thiosulfate quench. Generated two-stage process flow diagram and parameter comparison against the published flow protocol, including residence times, reactor volumes, total flow rate, and the unresolved mixing-criterion warning carried through to the final output.

FlowPilot generated a two-stage flow topology consisting of substrate and bromine pumps, a T-mixer, degassing, a first reactor, a quench pump, a second mixer, a quench/contact reactor, and product collection. The selected design used MeCN at 25 °C, no BPR, and a two-stage process matching the reported literature structure. The first reactor was assigned $\tau = 2.2$ min and $V_R = 5.30$ mL, and the quench/contact reactor was assigned $\tau = 1.0$ min and $V_R = 2.51$ mL. The total flow rate was 2.5 mL/min, and the total residence time was 3.2 min.

The council log shows that the agents correctly recognized this as a mixing-dominated problem. The Designer noted that all 12 candidates carried a dual-criterion mixing warning, but none were immediately disqualified. Chemistry Agent preferred shorter τ values with 0.75 mm ID tubing to minimize side reactions and emphasized continuous N_2 atmosphere. Kinetics Agent found that all candidates achieved

adequate estimated conversion, with $X = 0.98$ -1.00 and an intensification factor of approximately 10 relative to the 30 min batch time. Fluidics Agent flagged the dominant unresolved issue: every candidate failed the mixing criterion ($r_{\text{mix}} > 0.20$), suggesting that smaller tubing ID or improved micromixing should be considered. Safety Agent found no thermal or pressure risk but required standard MeCN handling controls and continuous N_2 . The Orchestrator selected Candidate 11 with a combined score of 0.756, primarily because it had the best geometry score among otherwise closely ranked alternatives.

The literature comparison shows strong agreement in qualitative process design and residence-time scale. FlowPilot matched the solvent, temperature, two-stage design, and absence of BPR. Its total residence time of 3.2 min fell within the reported 2-4 min literature range. The main discrepancy was scale: the literature process used approximately 30 mL/min and approximately 80 mL reactor volume, whereas FlowPilot proposed 2.5 mL/min and 7.8 mL. This is a scale difference rather than a chemistry-design mismatch. The case also demonstrates why council logs are useful: even when the final design agrees with literature residence time, the system preserves the unresolved mixing concern as an explicit warning rather than hiding it behind the final score.

DISCUSSION

The hybrid architecture separates chemical judgment from engineering arithmetic

The results above demonstrate that decomposing flow process design into two distinct reasoning modes; chemical interpretation by LLM agents and quantitative computation by a deterministic calculator grounded in handbook rules; addresses a structural weakness of monolithic LLM approaches. When a single model must simultaneously decide on a reaction mechanism, estimate a residence time, compute a Reynolds number, and size a back-pressure regulator, the outputs are often internally inconsistent because no enforcement mechanism checks that each quantity follows from a shared set of physical assumptions.

The FlowPilot architecture removes this failure mode by construction: the nine-step calculator derives every engineering quantity from a common set of inputs (flow rate, diameter, solvent properties, temperature), and LLM agents are not permitted to override these values. The pre/post-council comparison in Figure 4b illustrates the practical consequence. The upstream candidate carried weak plug-flow quality and a low intensification factor; after council review, the design shifted to a smaller tubing diameter and shorter residence time, nearly tripling the normalized radar-area score (Figure 4b). The council did not merely rephrase the upstream proposal; it materially revised the operating point by applying domain-specific constraints that the upstream model had not enforced. The α -bromination case study further shows that the council preserves unresolved engineering concerns, such as the mixing-

criterion violation flagged by Fluidics Agent, as explicit warnings in the final output instead of absorbing them into a confident summary. This transparency is what distinguishes a co-pilot from a black-box recommendation engine.

Positioning relative to prior AI systems for chemistry

Several recent systems have applied LLMs to chemistry tasks: ChemCrow³⁴ augments an LLM with tool calls for property lookup and safety assessment but operates in the batch domain; Coscientist³⁸ demonstrated autonomous experiment execution but targets the laboratory loop rather than process design; El Agente³⁵ automates quantum-chemistry calculations at the electronic-structure layer; Zeng et al.³⁴ applied a multi-agent framework to process optimization but assumes an already-defined process; and Materealize³⁷ employs multi-agent deliberation for materials design but does not incorporate fluid-dynamics or heat-transfer calculations. Classical process simulators²² require a pre-specified flowsheet and do not reason about chemistry, while Bayesian optimizers^{24–28} refine known processes but cannot propose one. FlowPilot addresses the segment between a chemistry brief and a defensible first flow design: it reasons about mechanism, generates engineering calculations, and audits the result through domain-specialist agents, producing a starting point for downstream experimental optimization or rigorous simulation.

The council architecture contributes beyond raw model capability

The model-matrix benchmark provides direct evidence that the council adds value beyond single-model inference. Across the 4×4 matrix, the upstream model determined the design region (residence time, reactor volume, tubing diameter), while the council model primarily affected execution cost and corrective iterations. Even Gemma upstream runs, which produced highly variable initial designs, yielded validated outputs after council review, indicating that the council acts as a quality floor rather than a pass-through.

The candidate-budget experiments add a second dimension. Increasing the pool from $B = 1$ to $B = 24$ broadened the explored design space but did not produce monotonically better final designs; at high budgets, the council occasionally selected long-residence-time solutions. This is consistent with constrained search rather than unconstrained optimization. In practice, a moderate budget ($B = 6–12$) offers a reasonable trade-off between exploration breadth and design robustness.

Limitations and scope boundaries

Several limitations should be noted. First, FlowPilot proposes designs but does not execute experiments. Closed-loop self-driving laboratories that integrate flow synthesis with AI-guided experimentation⁴⁴ could in principle complete this validation step, and combining FlowPilot with such platforms is a natural next step; nonetheless, experimental validation by a chemist remains essential at the design-evaluation stage. The 1.6-fold Stage 1 residence-time overestimate in Case Study 1 illustrates the consequence of kinetic uncertainty: without experimentally measured rate constants, the system relies on LLM-estimated kinetic anchors that tend toward conservative values.

Second, the deterministic calculator currently computes residence times on a liquid-phase basis and does not model gas-phase holdup in gas-liquid multiphase stages. For processes involving gaseous co-feeds such as the aerobic oxidation in Case Study 1, the reported residence times and reactor volumes reflect the liquid flow contribution only. Incorporating gas-liquid slug-flow models⁴⁵ would improve the fidelity of multiphase designs.

Third, the engineering rule base (2,537 rules) is not uniformly dense across chemistry classes. Thermal synthesis and photoredox catalysis are well represented, but electrochemistry, biocatalysis, and gas-phase flow chemistry have sparser coverage, which may reduce audit-trail reliability in those domains.

Fourth, the current benchmarks involve single-reaction or two-stage processes. Complex multi-step telescopes, where yield losses accumulate across stages, have not been evaluated and would require extensions to the topology builder and inter-stage consistency checks.

Fifth, frontier commercial models (Claude, GPT-4o) produced more consistent designs than the local Gemma 4 model, which required rescue pathways. Defining an acceptable quality floor for fully self-hosted deployment remains an open question.

Outlook

The most immediate extension is a closed-loop workflow in which FlowPilot designs are tested experimentally and the results are fed back to update kinetic anchors, reducing the conservatism observed in Case Study 1 and connecting the co-pilot to existing Bayesian optimization tools for flow-condition refinement^{24–28}. Expanding the rule base to electrochemistry, biocatalysis, and gas-liquid classes would broaden coverage. Because the deterministic calculator already tracks reactor volume, solvent inventory, and energy-related parameters (residence time, temperature, pressure drop), sustainability metrics such as E-factor and process mass intensity can be computed directly from the design output, connecting AI-

assisted process design to quantitative green chemistry evaluation. Supporting longer multi-step reaction telescopes and automated scale-up estimation from laboratory to pilot scale are further directions that the modular architecture is designed to accommodate.

CONCLUSION

We have presented FlowPilot, to our knowledge the first multi-agent AI system that translates a chemistry brief into a defensible continuous flow process design. The architecture enforces a strict division of labor: LLM agents interpret chemistry and mechanism, a nine-step deterministic calculator owns all engineering quantities, a handbook-derived rule base of approximately 2,500 rules supplies the domain constraints that experienced flow chemists rely on, and a seven-agent Council audits candidate designs under bounded domain authority. This separation is not incidental but structural; it prevents the internal inconsistencies that arise when a single model must simultaneously generate Reynolds numbers, pressure drops, residence times, and safety assessments without a shared physical basis.

Benchmarking across a 4×4 model matrix and candidate-budget sweeps shows that the council contributes measurable design quality beyond model capability alone, acting as a quality floor that revises weak upstream proposals instead of passing them through. The two end-to-end case studies, a photoredox Giese addition with downstream aerobic oxidation and a thermal α -bromination with inline quench, demonstrate that FlowPilot can recover chemically meaningful process topologies and residence-time scales consistent with published flow processes, while preserving unresolved engineering concerns as explicit warnings rather than absorbing them into a confident summary.

The practical implication for the flow chemistry community is a shift in the design bottleneck. Translating a batch protocol into a first flow design currently requires a team of specialists iterating over weeks; FlowPilot compresses this to a co-pilot proposal generated in minutes, with every number traceable to an equation, rule, or chemistry judgment. By systematically favoring intensified conditions; shorter residence times, smaller reactor volumes, and reduced solvent inventories; the system aligns AI-assisted process design with the resource-efficiency goals that motivate the adoption of continuous flow manufacturing. The system does not replace experimental validation, which remains essential, but it provides an auditable starting point from which experimental refinement can begin.

More broadly, the hybrid pattern demonstrated here, LLM-driven interpretation coupled with deterministic computation and multi-agent cross-domain audit, is not specific to flow chemistry. Any engineering design problem that is currently solved by a small team of complementary specialists

iterating across coupled domains is a candidate for the same architectural approach. The modular design of FlowPilot is intended to make such extensions practical.

METHODS

A literature corpus of 464 flow-chemistry papers was assembled and annotated with structured metadata (reaction class, reactor type, material, residence time, flow rate, yield, and stream count). From this corpus and from established flow-chemistry handbooks^{20,21,32}, an engineering knowledge base of 2,537 rules was extracted and categorized by topic and severity.

The plan-aware retrieval pipeline first generates a chemistry plan from the user input and then constructs an expanded query containing reaction class, mechanism, photocatalyst family, and sensitivity flags. Retrieved candidates are reranked by a field-aware score that rewards mechanistic relevance over superficial semantic similarity.

The batch-to-flow translation workflow comprises chemistry interpretation, literature retrieval, deterministic engineering calculation, candidate generation, council review, and topology rendering. A computable IntensificationMandate prevents acceptance of designs that do not measurably improve on batch. The nine-step deterministic calculator owns all numerical quantities (Re , Pe , Da , ΔP , V_R , τ , IF , UA , pressure headroom); LLM agents are not permitted to override these values.

The council reviews a candidate pool under bounded domain authority. Each of the seven agents (Designer, Chemistry Agent, Kinetics Agent, Fluidics Agent, Safety Agent, Skeptic, Orchestrator) may only modify fields within its own domain. Revised candidates are recomputed deterministically before final scoring.

The model-matrix benchmark evaluated 16 upstream/council pairings across Claude, GPT-4o, GPT-4o-mini, and Gemma 4. The candidate-budget benchmark tested $B = 1, 6, 12$, and 24 with five repeats each. Two end-to-end case studies^{42,43} were compared against published flow processes on topology, operating conditions, and residence-time scale.

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Author Contributions

A.M. and G.-N.A. contributed equally to this work. A.M. and G.-N.A. conceived the project and designed the FlowPilot architecture. A.M. developed the upstream chemistry-interpretation agent, the plan-aware retrieval pipeline, and the nine-step deterministic engineering calculator. G.-N.A. constructed the engineering rule base, implemented the Council and orchestration logic, and performed the model-matrix and candidate-budget benchmarks. Both authors designed and analyzed the case studies. A.M. and G.-N.A. authored the manuscript. Both authors read and approved the final manuscript.

Notes

Availability of data and code

Source code and data will be provided to peer reviewers via GitHub during the review process and publicly released upon publication of this work.

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ABBREVIATIONS

BPR, back-pressure regulator; Da, Damköhler number; FEP, fluorinated ethylene propylene; IF, intensification factor; LLM, large language model; MFC, mass flow controller; Pe, Péclet number; PFA, perfluoroalkoxy; PFD, process flow diagram; PTFE, polytetrafluoroethylene; RAG, retrieval-augmented generation; Re, Reynolds number; STY, space-time yield.

REFERENCES

- (1) Rossetti, I. Flow Chemistry: New Concepts from Batch to Continuous Organic Chemistry. *Ind. Chem.* **2016**, 2, e102. <https://doi.org/10.4172/2469-9764.1000e102>.

- (2) Rossetti, I.; Compagnoni, M. Chemical Reaction Engineering, Process Design and Scale-up Issues at the Frontier of Synthesis: Flow Chemistry. *Chem. Eng. J.* **2016**, *296*, 56–70. <https://doi.org/10.1016/j.cej.2016.02.119>.
- (3) Feng, K.; Zhu, Y.-Y.; Gu, S.; Long, J.; Wang, H.-F. Typical Cases of Continuous Flow Chemistry in Pharmaceutical Synthesis in 2023-2024. *Asian J. Org. Chem.* **2025**, *14*, e202500132. <https://doi.org/10.1002/ajoc.202500132>.
- (4) Wegner, J.; Ceylan, S.; Kirschning, A. Flow Chemistry – A Key Enabling Technology for (Multistep) Organic Synthesis. *Adv. Synth. Catal.* **2012**, *354*, 17–57. <https://doi.org/10.1002/adsc.201100584>.
- (5) Cambié, D.; Bottecchia, C.; Straathof, N.; Hessel, V.; Noël, T. Applications of Continuous-Flow Photochemistry in Organic Synthesis, Material Science, and Water Treatment. *Chem. Rev.* **2016**, *116*(17), 10276–10341. <https://doi.org/10.1021/acs.chemrev.5b00707>.
- (6) Padoin, N.; Matiazzo, T.; Riella, H.; Soares, C. A Perspective on the Past, the Present, and the Future of Computational Fluid Dynamics (CFD) in Flow Chemistry. *J. Flow Chem.* **2024**, *14*, 239–256. <https://doi.org/10.1007/s41981-024-00313-4>.
- (7) Movsisyan, M.; Delbeke, E.; Berton, J.; Battilocchio, C.; Ley, S.; Stevens, C. Taming Hazardous Chemistry by Continuous Flow Technology. *Chem. Soc. Rev.* **2016**, *45* (18), 4892–4928. <https://doi.org/10.1039/c5cs00902b>.
- (8) Yasukawa, T.; Kobayashi, S. Translation of Batch to Continuous Flow in Photoredox Reactions. *ACS Central Science*. July 16, 2021, pp 1099–1101. <https://doi.org/10.1021/acscentsci.1c00711>.
- (9) Noël, T.; Cao, Y.; Laudadio, G. The Fundamentals Behind the Use of Flow Reactors in Electrochemistry. *Acc. Chem. Res.* **2019**, *52*, 2858–2869. <https://doi.org/10.1021/acs.accounts.9b00412>.
- (10) Gemoets, H.; Su, Y.; Shang, M.; Hessel, V.; Luque, R.; Noël, T. Liquid Phase Oxidation Chemistry in Continuous-Flow Microreactors. *Chem. Soc. Rev.* **2016**, *45* (1), 83–117. <https://doi.org/10.1039/c5cs00447k>.
- (11) Plouffe, P.; Macchi, A.; Roberge, D. From Batch to Continuous Chemical Synthesis—A Toolbox Approach. *Org. Process Res. Dev.* **2014**, *18*, 1286–1294. <https://doi.org/10.1021/op5001918>.

- (12) Ingham, R. J.; Battilocchio, C.; Fitzpatrick, D. E.; Sliwinski, E.; Hawkins, J. M.; Ley, S. V. A Systems Approach Towards an Intelligent and Self-Controlling Platform for Integrated Continuous Reaction Sequences. *Angew. Chem. Int. Ed.* **2015**, *54* (1), 144–148. <https://doi.org/10.1002/anie.201409356>.
- (13) Adamo, A.; Beingessner, R. L.; Behnam, M.; Chen, J.; Jamison, T. F.; Jensen, K. F.; Monbaliu, J.-C. M.; Myerson, A. S.; Revalor, E. M.; Snead, D. R.; Stelzer, T.; Weeranoppanant, N.; Wong, S. Y.; Zhang, P. On-Demand Continuous-Flow Production of Pharmaceuticals in a Compact, Reconfigurable System. *Science* **2016**, *352* (6281), 61–67. <https://doi.org/10.1126/science.aaf1337>.
- (14) Holtze, C.; Boehling, R. Batch or Flow Chemistry? – A Current Industrial Opinion on Process Selection. *Curr. Opin. Chem. Eng.* **2022**, *36*, 100798. <https://doi.org/10.1016/j.coche.2022.100798>.
- (15) Glasnov, T. N.; Kappe, C. O. The Microwave-to-Flow Paradigm: Translating High-Temperature Batch Microwave Chemistry to Scalable Continuous-Flow Processes. *Chem. Eur. J.* **2011**, *17* (43), 11956–11968. <https://doi.org/10.1002/chem.201102065>.
- (16) Zhang, J.; Wang, K.; Teixeira, A.; Jensen, K.; Luo, G. Design and Scaling Up of Microchemical Systems: A Review. *Annu. Rev. Chem. Biomol. Eng.* **2017**, *8*, 285–305. <https://doi.org/10.1146/annurev-chembioeng-060816-101443>.
- (17) An, H.; Li, A.; Sasmito, A. P.; Kurnia, J. C.; Jangam, S. V.; Mujumdar, A. S. Computational Fluid Dynamics (CFD) Analysis of Micro-Reactor Performance: Effect of Various Configurations. *Chem. Eng. Sci.* **2012**, *75*, 85–95. <https://doi.org/10.1016/j.ces.2012.03.004>.
- (18) Ripken, R. M.; Wood, J. A.; Schlautmann, S.; Günther, A.; Gardeniers, H. J. G. E.; Le Gac, S. Towards Controlled Bubble Nucleation in Microreactors for Enhanced Mass Transport. *React. Chem. Eng.* **2021**, *6* (10), 1869–1877. <https://doi.org/10.1039/D1RE00092F>.
- (19) Mielke, E.; Plouffe, P.; Koushik, N.; Eyholzer, M.; Gottsponer, M.; Kockmann, N.; Macchi, A.; Roberge, D. M. Local and Overall Heat Transfer of Exothermic Reactions in Microreactor Systems. *React. Chem. Eng.* **2017**, *2* (5), 763–775. <https://doi.org/10.1039/C7RE00085E>.
- (20) Capaldo, L.; Wen, Z.; Noël, T. A Field Guide to Flow Chemistry for Synthetic Organic Chemists. *Chem. Sci.* **2023**, *14* (16), 4230–4247. <https://doi.org/10.1039/D3SC00992K>.

- (21) Plutschack, M. B.; Pieber, B.; Gilmore, K.; Seeberger, P. H. The Hitchhiker's Guide to Flow Chemistry. *Chem. Rev.* **2017**, *117* (18), 11796–11893. <https://doi.org/10.1021/acs.chemrev.7b00183>.
- (22) Haydary, J. *Chemical Process Design and Simulation*; Wiley: **2018**. <https://doi.org/10.1002/9781119311478>.
- (23) Reynoso-Donzelli, S.; Ricardez-Sandoval, L. A Reinforcement Learning Approach with Masked Agents for Chemical Process Flowsheet Design. *AIChE J.* **2025**, *71* (1), e18584. <https://doi.org/10.1002/aic.18584>.
- (24) Mottafegh, A.; Joo, J.-U.; Na, G.-S.; Sharma, V.; Kim, D.-P. AI-Assisted Autonomous Manufacturing of Tailored Drug-Loaded Nanoparticles by Multi-Step Continuous-Flow Platform. *Chem. Eng. J.* **2024**, *500*, 157454. <https://doi.org/10.1016/j.cej.2024.157454>.
- (25) Dunlap, J.; Ethier, J.; Putnam-Neeb, A.; Iyer, S.; Luo, S.-X. L.; Feng, H.; Torres, J. G.; Doyle, A.; Swager, T.; Vaia, R.; Mirau, P.; Crouse, C.; Baldwin, L. Continuous Flow Synthesis of Pyridinium Salts Accelerated by Multi-Objective Bayesian Optimization with Active Learning. *Chem. Sci.* **2023**, *14*, 8061–8069. <https://doi.org/10.1039/d3sc01303k>.
- (26) Ahn, G.-N.; Kang, J.-H.; Lee, H.-J.; Park, B. E.; Kwon, M.; Na, G.-S.; Kim, H.; Seo, D.-H.; Kim, D.-P. Exploring Ultrafast Flow Chemistry by Autonomous Self-Optimizing Platform. *Chem. Eng. J.* **2023**, *453*, 139707. <https://doi.org/10.1016/j.cej.2022.139707>.
- (27) Mottafegh, A.; Ahn, G.-N.; Kim, D.-P. Meta Optimization Based on Real-Time Benchmarking of Multiple Surrogate Models for Autonomous Flow Synthesis. *Lab. Chip* **2023**, *23* (6), 1613–1621. <https://doi.org/10.1039/D2LC00938B>.
- (28) Jorayev, P.; Soritz, S.; Sung, S.; Jeraal, M.; Russo, D.; Barthelme, A.; Toussaint, F.; Gaunt, M.; Lapkin, A. Machine Learning-Driven Optimization of Continuous-Flow Photoredox Amine Synthesis. *Org. Process Res. Dev.* **2025**, *29*, 1411–1422. <https://doi.org/10.1021/acs.oprd.4c00533>.
- (29) Tu, Z.; Choure, S.; Fong, M. H.; Roh, J.; Levin, I.; Yu, K.; Joung, J.; Morgan, N.; Li, S.-C.; Sun, X.; Lin, H.; Murnin, M.; Liles, J.; Struble, T.; Fortunato, M.; Liu, M.; Green, W.; Jensen, K.; Coley, C. ASKCOS: Open-Source, Data-Driven Synthesis Planning. *Acc. Chem. Res.* **2025**, *58* (11), 1764–1775. <https://doi.org/10.1021/acs.accounts.5c00155>.

- (30) Coley, C.; Green, W.; Jensen, K. Machine Learning in Computer-Aided Synthesis Planning. *Acc. Chem. Res.* **2018**, *51* (5), 1281–1289. <https://doi.org/10.1021/acs.accounts.8b00087>.
- (31) Hartog, P.; Westerlund, A.; Tetko, I.; Genheden, S. Investigations into the Efficiency of Computer-Aided Synthesis Planning. *J. Chem. Inf. Model.* **2025**, *65* (4), 1771–1781. <https://doi.org/10.1021/acs.jcim.4c01821>.
- (32) Guidi, M.; Seeberger, P. H.; Gilmore, K. How to Approach Flow Chemistry. *Chem. Soc. Rev.* **2020**, *49* (24), 8910–8932. <https://doi.org/10.1039/C9CS00832B>.
- (33) Ramos, M. C.; Collison, C. J.; White, A. D. A Review of Large Language Models and Autonomous Agents in Chemistry. arXiv November 14, 2024. <https://doi.org/10.48550/arXiv.2407.01603>.
- (34) M. Bran, A.; Cox, S.; Schilter, O.; Baldassari, C.; White, A. D.; Schwaller, P. Augmenting Large Language Models with Chemistry Tools. *Nat. Mach. Intell.* **2024**, *6* (5), 525–535. <https://doi.org/10.1038/s42256-024-00832-8>.
- (35) Zou, Y.; Cheng, A. H.; Aldossary, A.; Bai, J.; Leong, S. X.; Campos-Gonzalez-Angulo, J. A.; Choi, C.; Ser, C. T.; Tom, G.; Wang, A.; Zhang, Z.; Yakavets, I.; Hao, H.; Crebolder, C.; Bernales, V.; Aspuru-Guzik, A. El Agente: An Autonomous Agent for Quantum Chemistry. *Matter* **2025**, *8* (7), 102263. <https://doi.org/10.1016/j.matt.2025.102263>.
- (36) Zeng, T.; Badrinarayanan, S.; Ock, J.; Lai, C.-K.; Barati Farimani, A. LLM-Guided Chemical Process Optimization with a Multi-Agent Approach. *Mach. Learn. Sci. Technol.* **2025**, *6* (4), 045067. <https://doi.org/10.1088/2632-2153/ae2382>.
- (37) Kim, S.; Choi, J.; Jang, K.; Park, J.; Bernales, V.; Aspuru-Guzik, A.; Jung, Y. Materealize: A Multi-Agent Deliberation System for End-to-End Material Design and Synthesis. arXiv January 22, 2026. <https://doi.org/10.48550/arXiv.2601.15743>.
- (38) Boiko, D. A.; MacKnight, R.; Kline, B.; Gomes, G. Autonomous Chemical Research with Large Language Models. *Nature* **2023**, *624* (7992), 570–578. <https://doi.org/10.1038/s41586-023-06792-0>.
- (39) Guo, X.; Li, Y.; Kong, X.; Jiang, Y.; Zhao, X.; Gong, Z.; Zhang, Y.; Li, D.; Sang, T.; Zhu, B.; Jun, G.; Huang, Y.; Liu, Y.; Xue, Y.; Kundu, R. D.; Lim, Q. J.; Zhao, Y.; Granger, L. A.; Younis, M. B.; Keivan, D.; Sabharwal, N.; Sinha, S.; Agarwal, P.; Vandyck, K.; Mai, H.; Wang, Z.; Venkatesh, A.; Barik, A.; Yang, J.; Yue, C.; He, J.; Wang, L.; Xu, L.; Chen, H.; Wang, J.; Xu, L.; Shetty, R.; Guo, Z.; Song, D.; Jha, M.; Liang, W.; Yan, W.; Zhang, B.; Karnoor, S. B.; Zhang, J.; Pandya, R.; Gong,

- X.; Ganesh, M. B.; Shi, F.; Xu, R.; Zhang, Y.; Ouyang, Y.; Qin, L.; Rosenbaum, E.; Snyder, C.; Seiler, P.; Dullerud, G.; Zhang, X. S.; Cheng, Z.; Hanumolu, P. K.; Huang, J.; Kulkarni, M.; Namazifar, M.; Zhang, H.; Hu, B. Toward Engineering AGI: Benchmarking the Engineering Design Capabilities of LLMs. arXiv September 21, 2025. <https://doi.org/10.48550/arXiv.2509.16204>.
- (40) Norheim, J. J.; Rebentisch, E.; Xiao, D.; Draeger, L.; Kerbrat, A.; de Weck, O. L. Challenges in Applying Large Language Models to Requirements Engineering Tasks. *Des. Sci.* **2024**, *10*, e16. <https://doi.org/10.1017/dsj.2024.8>.
- (41) Ahn, G.-N.; Shin, J.-A.; Lee, S. Sustainable Continuous Synthesis of 3,5-Disubstituted Isoxazoles: Development of an Integrated Reaction–Separation–Recovery System for Deep Eutectic Solvents. *ACS Sustainable Chem. Eng.* **2025**, *13*, 6589–6597. <https://doi.org/10.1021/acssuschemeng.5c00580>.
- (42) Park, J.; Kim, S. H.; Cho, J.-Y.; Atriardi, S. R.; Kim, J.-Y.; Mardhiyah, H.; Park, B. Y.; Woo, S. K. Batch and Flow Synthesis of Sulfides and Sulfoxides Using Green Solvents and Oxidant through Visible-Light Photocatalysis. *Green Chem.* **2025**, *27*, 3284–3292. <https://doi.org/10.1039/d4gc05769d>.
- (43) Lu, Y.; Yan, Z.; Jiang, Y.; He, H.; Zhou, J.; Gao, S.; Xing, D. α -Bromination of Aldehydes by Continuous Flow Chemistry and Its Application to the Flow Synthesis of 2-Aminothiazoles. *Org. Process Res. Dev.* **2025**, *29*, 2047–2055. <https://doi.org/10.1021/acs.oprd.5c00076>.
- (44) Coley, C. W.; Thomas III, D. A.; Lummiss, J. A. M.; Jaworski, J. N.; Breen, C. P.; Schultz, V.; Hart, T.; Fishman, J. S.; Rogers, L.; Gao, H.; Hicklin, R. W.; Plehiers, P. P.; Byington, J.; Piotti, J. S.; Green, W. H.; Hart, A. J.; Jamison, T. F.; Jensen, K. F. A Robotic Platform for Flow Synthesis of Organic Compounds Informed by AI Planning. *Science* **2019**, *365* (6453), eaax1566. <https://doi.org/10.1126/science.aax1566>.
- (45) Mallia, C. J.; Baxendale, I. R. The Use of Gases in Flow Synthesis. *Org. Process Res. Dev.* **2016**, *20* (2), 327–360. <https://doi.org/10.1021/acs.oprd.5b00222>.